Documentation

Supplementary raw data on cell chaining in an attenuated Listeria monocytogenes

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Abstract
The dataset "Supplementary raw data on cell chaining in an attenuated Listeria monocytogenes" provides the raw images and data for the manuscript "On the role of cell chaining in the attenuation of a Listeria monocytogenes divIVA mutant" (https://doi.org/10.64898/2026.01.05.697648).

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Information on the dataset and context of origin

The dataset "Supplementary raw data on cell chaining in an attenuated Listeria monocytogenes" provides the raw images and data, which form the basis for the figures of the manuscript "On the role of cell chaining in the attenuation of a Listeria monocytogenes divIVA mutant". In-depth information on the background of the work, on methodology and on the analysis of the results can be found in

> Wamp, S., Holland, G., Rismondo. J., and Halbedel, S. (2026). On the role of cell chaining in the attenuation of a Listeria monocytogenes divIVA mutant. bioRxiv. https://doi.org/10.64898/2026.01.05.697648

Administrative and organizational information

The publicaton "Supplementary raw data on cell chaining in an attenuated Listeria monocytogenes", is a joined work of the Consultant Laboratory for Listeria in Unit 11| Enteropathogenic Bacteria and Legionella of the Robert Koch Institute, the Unit ZBS 4| Advanced Light and Electron Microscopy and the Institute for Medical Microbiology and Hospital Hygiene at the Otto von Guericke University. Questions regarding the content of the data can be addressed directly to the corresponding author Sven Halbedel (halbedels@rki.de).
The publication of the data as well as the quality management of the (meta-)data is done by the unit MF 4 | Domain Specific Data and Research Data Management. Questions regarding data management and the publication infrastructure can be directed to the Open Data Team of the unit MF4 at OpenData@rki.de.

Overview over the *L. monocytogenes* strains used for this manuscript

The following strains were used for this manuscript. Detailed methodology on their generation can be found in the methods section of the manuscript (https://doi.org/10.64898/2026.01.05.697648).

Strain name	Description
EGD-e (wt)	Wildtype strain *L. monocytogenes*, used as control
LMSW49 (i*mreB*)	MreB depletion strain, MreB expression can be induced upon IPTG-treatment
LMJR183 (i*ezrA*)	EzrA depletion strain, EzrA expression can be induced upon IPTG-treatment
LMS2 (Δ*divIVA*)	divIVA deletion strain
LMSW207 (Δ*divIVA* i*mreB*)	divIVA deletion and MreB depletion strain, MreB expression can be induced upon IPTG-treatment
LMSW208 (Δ*divIVA* i*ezrA*)	divIVA deletion and EzrA depletion strain, EzrA expression can be induced upon IPTG-treatment
BUG2214 (Δ*prfA*)	prfA deletion strain
LMS30 (i*divIVA*)	divIVA depletion strain, divIVA expression can be induced upon IPTG-

Strain name	Description
	treatment
LMJD20 (wt)	DsRed producing *L. monocytogenes* strain, used as control
LMSF2 (Δ*actA*)	DsRed producing, actA deletion strain
LMSW205 (Δ*divIVA*)	DsRed producing, divIVA deletion strain
LMSW202 (*iezrA*)	DsRed producing, EzrA depletion strain, EzrA expression can be induced upon IPTG-treatment
LMSW203 (*imreB*)	DsRed producing, MreB depletion strain, MreB expression can be induced upon IPTG-treatment
BUG3057 (*prfA**)	*L. monocytogenes* strain that encodes a PrfA variant that is constitutively active causing constitutive overexpression of PrfA-dependent virulence genes
LMSW251 (Δ*divIVA prfA**)	divIVA deletion strain that encodes a PrfA variant that is constitutively active causing constitutive overexpression of PrfA-dependent virulence genes
LMSW222 (Δ*divIVA secA2 M95I*)	divIVA deletion strain with secA2 mutation (M95I)
LMSW223 (Δ*divIVA secA2 T123A*)	divIVA deletion strain with secA2 mutation (T123A)

Content and structure of the raw supplemented raw images and data TBD Version A per figure

The raw data and relevant calculations that form the basis for the manuscript's figures are provided as .xlsx files.

The raw microscopy images, as well as uncropped images from a western blot, a gel electrophoresis and culture plates are provided in the "Images" subfolder as .tif files.
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Figure	Description	File
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Figure	Description	File
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Metadata

TODO: Footer here .....

Content and structure of the raw supplemented raw images and data TBD Version B data and images separated

Data files

The raw data and relevant calculations that form the basis for the manuscript's figures are provided as .xlsx files. The files are named according to the figures for which they form the basis.

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Image files

The raw microscopy images, as well as uncropped images from a western blot, a gel electrophoresis and culture plates are provided in the “Images” subfolder as .tif files.
> Images/

The files are named according to the figures for which they form the basis.

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Figure S1	Scanning electron microscopy image of LMJR183 (i*ezrA*)	Images/Figure_S1_ezrA_raw_image.tif
Figure S1	TODO: missing image, Scanning electron microscopy image of LMSW49 (i*mreB*)	Images/Figure_S1_mreB_raw_image.tif
Figure S3	Fluorescence microscopy image of LMJD20 (wt, DAPI) wt_DAPI	Images/Figure_S3_wt_DAPI_raw_image.tif
Figure S3	Fluorescence microscopy image of LMJD20 (wt, phase contrast)	Images/Figure_S3_wt_pc_raw_image.tif
Figure S3	Fluorescence microscopy image of LMJD20 (wt, TRITC)	Images/Figure_S3_wt_TRITC_raw_image.tif
Figure S3	Fluorescence microscopy image of LMSW202 (i*ezrA*, DAPI)	Images/Figure_S3_ezrA_DAPI_raw_image.tif
Figure S3	TODO: missing image, Fluorescence microscopy image of LMSW202 (i*ezrA*, phase contrast)	Images/Figure_S3_ezrA_pc_raw_image.tif
Figure S3	Fluorescence microscopy image of LMSW202 (i*ezrA*, TRITC)	Images/Figure_S3_ezrA_TRITC_raw_image.tif

Figure	Description	File
Figure S3	Fluorescence microscopy image of LMSW203 (i*mreB*, DAPI)	Images/Figure_S3_mreB_DAPI_raw_image.tif
Figure S3	Fluorescence microscopy image of LMSW203 (i*mreB*, phase contrast)	Images/Figure_S3_mreB_pc_raw_image.tif
Figure S3	Fluorescence microscopy image of LMSW203 (i*mreB*, TRITC)	Images/Figure_S3_mreB_TRITC_raw_image.tif
Figure S3	Fluorescence microscopy image of LMSW205 (Δ*divIVA*, DAPI)	Images/Figure_S3_divIVA_DAPI_raw_image.tif
Figure S3	Fluorescence microscopy image of LMSW205 (Δ*divIVA*, phase contrast)	Images/Figure_S3_divIVA_pc_raw_image.tif
Figure S3	Fluorescence microscopy image of LMSW205 (Δ*divIVA*, TRITC)	Images/Figure_S3_divIVA_TRITC_raw_image.tif
Figure S4	Phospholipolytic activity on egg-yolk agar of EGD-e (wt)	Images/Figure_S4_egg_yolk_agar_wt_raw_image.tif
Figure S4	Phospholipolytic activity on egg-yolk agar of LMS2 (Δ*divIVA*)	Images/Figure_S4_egg_yolk_agar_divIVA_raw_image.tif
Figure S4	Phospholipolytic activity on egg-yolk agar of BUG2214 (Δ*prfA*)	Images/Figure_S4_egg_yolk_agar_prfA_raw_image.tif
Figure S4	Phospholipolytic activity on egg-yolk agar of BUG3057 (*prfA**)	Images/Figure_S4_egg_yolk_agar_prfA-star_raw_image.tif
Figure S4	Phospholipolytic activity on egg-yolk agar of LMSW251 (Δ*divIVA prfA**)	Images/Figure_S4_egg_yolk_agar_divIVA-prfA-star_raw_image.tif
Figure S4	Hemolytic activity on sheep blood agar of EGD-e (wt), BUG2214 (Δ*prfA*) and BUG3057 (*prfA**)	Images/Figure_S4_hemolysis_wt_prfA_prfA-star_raw_image.tif
Figure S4	Hemolytic activity on sheep blood agar of LMS2 (Δ*divIVA*) and LMSW251 (Δ*divIVA prfA**)	Images/Figure_S4_hemolysis_divIVA_divIVA-prfA-star_raw_image.tif

Figure	Description	File
Figure S4	Swarming on soft agar of EGD-e (wt), BUG2214 (Δ*prfA*), BUG3057 (*prfA**), LMS2 (Δ*divIVA*) and LMSW251 (Δ*divIVA prfA**)	Images/Figure_S4_swarming_plate_raw_image.tif
Figure S4	Cellular morphology after nile red staining of EGD-e (wt)	Images/Figure_S4_nile_red_wt_raw_image.tif
Figure S4	Cellular morphology after nile red staining of LMS2 (Δ*divIVA*)	Images/Figure_S4_nile_red_divIVA_raw_image.tif
Figure S4	Cellular morphology after nile red staining of BUG2214 (Δ*prfA*)	Images/Figure_S4_nile_red_prfA_raw_image.tif
Figure S4	Cellular morphology after nile red staining of BUG3057 (*prfA**)	Images/Figure_S4_nile_red_prfA-star_raw_image.tif
Figure S4	Cellular morphology after nile red staining of LMSW251 (Δ*divIVA prfA**)	Images/Figure_S4_nile_red_divIVA-prfA-star_raw_image.tif

Metadata

To increase findability, the provided data are described with metadata. The Metadata are distributed to the relevant platforms via GitHub Actions. There is a specific metadata file for each platform; these are stored in the metadata folder:

> Metadata/

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"publication_date": "2024-06-19",
  "dates": [
    {
      "start": "2023-09-11T15:00:21+02:00",
      "end": "2023-09-11T15:00:21+02:00",
      "type": "Created",
      "description": "Date when the published data was created"
    }
  ],
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